

StatAna Summative 2 Reviewer: Modules 3 and 4

CS0073 Statistical Analysis and Modeling. Covers Module 3 (Predictive Analytics and Regression Modeling) and Module 4 (Prescriptive Analytics and Decision Optimization). This also serves as the M3/M4 half of the midterm reviewer (midterm covers M1 to M4; M1 and M2 are not included here because those slides are not available yet). The blue boxes are memory aids I made; they are not from the slides.

What Is Inside

- **Part 1, Module 3:** the four types of analytics, the predictive analytics process, the linear regression equation and the Sales and Ad Spending example, multiple regression, the training/validation/testing split, a full walkthrough of how to read an Excel regression output using the slide's own numbers, and the four forecast accuracy measures.
- **Part 2, Module 4:** prescriptive analytics and "what should we do?", the evolution of analytics, the five-step prescriptive process, key technologies, business applications by industry, decision-making benefits, implementation challenges and solutions, and future trends.
- A single one-page cram sheet covering both parts comes at the very end. Read it last, right before the exam.

Fast Memory Map

Topic	Hook
Four analytics types	Descriptive, Diagnostic, Predictive, Prescriptive = What, Why, Will, Should
Predictive process	Project design, Sampling, Exploration, then a cycle of Modification, Development, Validation
Regression equation	$y = \beta x + \alpha + \epsilon$; Y depends, X drives, Beta is the Rate, Alpha is the Base, Epsilon Evens it out
R Square	0.8937 means 89.37 percent of sales variation explained
ANOVA df (n=140, 3 predictors)	3, 136, 139
Significant predictors	TV and Radio significant ($p < 0.05$); Newspaper is not ($p = 0.6690$)
When p is low	H0 must go
Forecast error measures	MAE plain, MSE squared, RMSE un-squared, MAPE percent of actual Y
Prescriptive process	Data, Predict, Scenario, Optimize, Act (5 steps)
Prescriptive question	"What should we do?"

Part 1: Module 3, Predictive Analytics and Regression Modeling

The Four Types Of Analytics

There are four types of analytics, each answering a different question. Each stage builds on the previous one, with increasing business value and complexity.

Type	Question It Answers
Descriptive	What has happened, or what is happening now?
Diagnostic	Why did it happen?
Predictive	What will likely happen?
Prescriptive	What should we do?

Memory Aid

D-D-P-P in order of maturity: Descriptive, Diagnostic, Predictive, Prescriptive. Each one answers a harder question than the last: What, Why, Will, Should.

The Predictive Analytics Process

The process starts with three stages, then loops through a three-step cycle until the best model is selected:

1. **Project Design:** kickoff meeting, understand the modeling objective, define acceptance criteria, document data and deployment requirements.
2. **Data Sampling:** data extraction, apply filters and exclusions, identify external data sources.
3. **Data Exploration:** exploratory data analysis (EDA), identify data dependencies and correlations, identify trends or anomalies in the data.
4. **Data Modification:** data cleaning, data augmentation and transformation, feature selection.
5. **Model Development:** apply different modeling techniques and select the final methodology.
6. **Model Validation:** model performance review, feedback based on business knowledge and inputs from subject matter experts (SMEs).

Data Modification, Model Development, and Model Validation form a cycle that loops back into Project Design and best model selection, repeating until the final model is chosen.

Memory Aid

P-S-E, then a cycle of M-D-V: Project design, Sampling, Exploration, then Modification, Development, Validation. The last three steps are not one-and-done: they loop back into Project Design until the best model is selected.

Defining A Linear Regression Problem

A linear regression problem is defined by two questions: **what is it?** (a relationship between two things) and **how is it used?** (one thing, A, causing an effect on another thing, B).

The Simple Linear Regression Equation

The core formula is:

Formula

$$y = \beta x + \alpha + \varepsilon$$

Term	Name	Meaning
y	Dependent variable	The value to be predicted
x	Independent variable	The value driving the prediction
β (beta)	Beta coefficient	The rate multiplied to x
α (alpha)	Alpha intercept	The baseline figure for y
ε (epsilon)	Error term	The balancing figure

Memory Aid

Y depends, X drives, Beta is the Rate, Alpha is the Base, Epsilon Evens it out.

The Sales And Ad Spending Example

The slide maps the formula onto a real case: predicting sales from advertising spend.

Formula Term	In This Example
y, dependent variable	Sales
x, independent variable	Ad Spending
β, beta coefficient	Sales Sensitivity
α, alpha intercept	Baseline Sales

Why The Error Term Exists

The slides give two reasons for including ε :

- To account for **unexplained variability** in the dependent variable from other relevant independent variables that may not have been included in the model.
- To capture **measurement error** in both the dependent and independent variables.

Memory Aid

Error term = everything the model missed. Missing variables, plus measurement mistakes.

Multiple Linear Regression

You can have more than one predictor variable. The formula extends from one x to many:

Formula

$$y = \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n + \alpha + \varepsilon$$

This is the same equation as simple linear regression, just with more predictors (x_1 through x_n), each with its own beta coefficient.

Splitting The Dataset

Before training a model, the original data is split so the model can be evaluated fairly:

1. **Original Data** is split into **Training Data** and **Testing Data**.
2. **Training Data** is split again into **Training** and **Validation**.

Split	Purpose
Training data	Trains the algorithm
Validation data	Used to tune and evaluate the model while it is being built
Testing data	The final performance evaluation of the finished model

Memory Aid

Train to learn, Validate to tune, Test to grade. The testing set is only touched at the very end.

How To Read An Excel Regression Output

The slides show a regression of sales on three ad spend predictors (TV, radio, newspaper), with a sample size of **n = 140**. Read an Excel regression output in this order: Regression Statistics, ANOVA, then the coefficients (t-test) table.

Step 1: Regression Statistics (R Square)

Statistic	Value
Multiple R	0.94536
R Square	0.893710
Adjusted R Square	0.891366
Standard Error	1.736514
Observations	140

R Square (the coefficient of multiple determination, R²) is a goodness-of-fit measure. It is normally expressed as a percentage and is interpreted as the amount of variability in the response explained by the independent variables. R² is a figure of merit: the higher the R², the better the model explains the variation in the response using the set of predictors. Here, **R Square 0.8937 means 89.37 percent of the variation in sales can be explained by TV, radio, and newspaper ad spend.**

Watch Out

R² almost always goes up when you add more predictors, even useless ones. **Adjusted R Square** corrects for the number of predictors, so it is the fairer number to compare between models with different numbers of predictors. On this slide Adjusted R Square (0.891366) is close to but slightly below R Square (0.893710), which is expected.

Step 2: ANOVA Table

The ANOVA (Analysis of Variance) table is a decomposition of the total variation in the response into explained (pattern) and unexplained (error) parts. The explained variability is the amount of variation in the response variable that may be attributed to the predictors explicitly stated in the model. The unexplained variability is the amount of variation attributed to random error.

Source	df	SS	MS	F	Significance F
Regression	3	3,448.264191	1,149.421397	381.1737161	5.6038E-66
Residual	136	410.1051657	3.015479159		
Total	139	3,858.369357			

SS (Sum of Squares): 3,448.26 is the variation in sales explained by the three predictors; 410.11 is the unexplained variation; the two sum to 3,858.37 (Total). A good fit shows the Regression SS much larger than the Residual SS, which is the case here.

df (degrees of freedom), for n = 140 and 3 predictors: Regression df = number of regression parameters minus one = **3**. Residual df = sample size minus number of regression parameters = **136**. Total df = Regression df + Residual df = **139**.

MS (Mean Squares) is each SS divided by its own df. MS has no physical meaning on its own; it exists to compute the F statistic.

F and Significance F: the F-test determines if the regression is meaningful for the data at hand. When the p-value (Significance F) is small, at least one predictor is significant. The rule of thumb is the p-value is low if it is less than the alpha significance level (usually 0.05). Here Significance F is 5.6038E-66, an extremely small number, so the regression as a whole is significant.

Memory Aid

"When p is low, H0 must go!" A small p-value means you reject the null hypothesis (H0), which for the ANOVA F-test is "none of the predictors matter."

Watch Out

df formula is a frequent trap. Regression df = (number of parameters) minus 1 = 3. Residual df = n minus (number of parameters) = 140 - 4 = 136. Total df = 3 + 136 = 139. Do not confuse "number of parameters" (4: intercept plus 3 predictors) with "number of predictors" (3).

Step 3: t-Tests Per Predictor

The t-test helps in assessing if an individual predictor is significant.

Predictor	Coefficient	SE	t	p-value
Intercept	3.045142209	0.391309656	7.781924521	1.60861E-12
TV Ads (in 000s)	0.047048681	0.001701367	27.6534582	1.0917E-57
Radio Ads (in 000s)	0.179682989	0.010782334	16.6645727	1.16107E-34
Newspaper Ads (in 000s)	-0.003005565	0.007014537	-0.42847661	0.668982012

- **TV ad spend:** $p < 0.05$, so TV ad spend is a **significant** predictor of sales.
- **Radio ad spend:** $p < 0.05$, so radio ad spend is a **significant** predictor of sales.
- **Newspaper ad spend:** $p > 0.05$ (0.668982012), so newspaper ad spend is an **insignificant** predictor of sales.

Memory Aid

TV and Radio pass (low p, significant). Newspaper fails (high p, insignificant). Newspaper is the one to remember as the "odd one out" on this slide.

Watch Out

Newspaper being insignificant is a favorite trap. Its p-value (0.668982) is far above 0.05,

and its coefficient is even negative. Do not assume all three predictors in a multiple regression are automatically useful just because the overall model (ANOVA) is significant.

Step 4: Interpreting The Coefficients

Each coefficient is read **holding the other predictors constant**. The slide gives the fitted model as:

Fitted Model (as printed on the slide)
 $\text{Sales} = 2.98 + 0.047 \times \text{TV} + 0.178 \times \text{Radio}$

- **Intercept, 2.98:** the average sales if TV, radio, and newspaper ad spends are all 0.
- **0.047 (TV):** an estimated increase of 0.047 million (P47,000) in average sales for every P1,000 spent on TV ads, **holding radio and newspaper constant**.
- **0.178 (Radio):** an estimated increase of 0.178 million (P178,000) in average sales for every P1,000 spent on radio ads, **holding TV and newspaper constant**.

Watch Out

The slides disagree on the exact coefficients. The p26 t-test table (the full model with all three predictors) gives Intercept 3.045142209, TV 0.047048681, Radio 0.179682989, Newspaper -0.003005565. The p28 fitted model slide instead prints **Sales = 2.98 + 0.047 x TV + 0.178 x Radio**, dropping newspaper entirely and rounding the other two numbers (2.98 vs 3.045, 0.178 vs 0.1797). The most likely explanation is the p28 model was refit without the insignificant newspaper predictor. If an exam question asks for "the fitted model," quote the p28 numbers (2.98, 0.047, 0.178). If it asks for coefficients, SE, t, or p from the t-test table, use the p26 numbers.

Step 5: Forecast Accuracy Measures

Once a model makes predictions, forecast error for each observation is $e_i = Y - \hat{Y}$ (actual minus predicted). Four measures summarize how accurate the model is:

Measure	Formula	What It Means
MAE (Mean Absolute Error)	$(1/n) \text{ sum of } e_i $	Average size of the errors, ignoring direction (over- or under-prediction)
MSE (Mean Squared Error)	$(1/n) \text{ sum of } e_i \text{ squared}$	Average of the squared errors; squaring punishes big misses more than small ones
RMSE (Root Mean Squared Error)	square root of MSE	MSE brought back to the original units of y, so it is easier to interpret than MSE
MAPE (Mean Absolute Percentage Error)	$(1/n) \text{ sum of } (e_i / Y), \text{ as a percentage}$	Average error as a percentage of the actual value, so it is comparable across different scales

Memory Aid

MAE averages the absolute errors, in the original units. MSE averages the squared errors, in squared units, so it cannot be compared directly to MAE or RMSE. RMSE is the square root of MSE, back in the original units. MAPE expresses the average absolute error as a percentage of the actual value.

Watch Out

MAPE divides by the actual Y, not the predicted value. Mixing this up is a common exam trap.

Practice: Read This Output

Using only the numbers already on these slides: a regression of sales on TV, radio, and newspaper ad spend, $n = 140$, gives $R^2 = 0.8937$, ANOVA $df = 3, 136, 139$ with Significance $F = 5.6038E-66$, and t-test p-values of $1.09E-57$ (TV), $1.16E-34$ (Radio), and 0.6690 (Newspaper).

1. **How much of the variation in sales does the model explain?** 89.37 percent ($R^2 = 0.8937$).
2. **Is the overall regression significant?** Yes. Significance $F (5.6038E-66)$ is far below 0.05.
3. **Which predictors are individually significant at the 0.05 level?** TV and Radio ($p < 0.05$). Newspaper is not ($p = 0.6690$, above 0.05).
4. **What are the Regression, Residual, and Total degrees of freedom?** 3, 136, and 139, matching 3 predictors and $n = 140$.

Part 2: Module 4, Prescriptive Analytics and Decision Optimization

Introduction To Prescriptive Analytics

Prescriptive analytics is the most advanced form of business analytics. It goes beyond describing what happened (descriptive), explaining why it happened (diagnostic), or predicting what will happen (predictive). Prescriptive analytics recommends actions to achieve desired outcomes and tells you the implications of each decision option. **It answers the question: "What should we do?"** By leveraging advanced algorithms, machine learning, and optimization techniques, prescriptive analytics enables data-driven decision making at scale.

Memory Aid

Descriptive, Diagnostic, Predictive, Prescriptive = What, Why, Will, Should. Prescriptive is the only one that tells you what action to take.

The Evolution Of Analytics

Stage	Question
Descriptive Analytics	"What happened?" Examines historical data to understand past performance.
Diagnostic Analytics	"Why did it happen?" Identifies causes and relationships in data.
Predictive Analytics	"What will happen?" Forecasts future outcomes based on patterns.
Prescriptive Analytics	"What should we do?" Recommends actions to optimize outcomes.

Each stage builds upon the previous, with increasing business value and complexity. Prescriptive analytics represents the pinnacle of data-driven decision making.

How Prescriptive Analytics Works: The Five-Step Process

1. **Data Collection and Integration:** gather relevant historical and real-time data from multiple sources.
2. **Predictive Modeling:** create models to forecast potential future outcomes.
3. **Scenario Analysis:** evaluate multiple possible scenarios and their implications.
4. **Optimization:** apply algorithms to determine the best course of action.
5. **Action Recommendation:** deliver actionable insights to decision-makers.

This is an iterative process that continuously improves as more data becomes available and outcomes are measured.

Memory Aid

D-P-S-O-A: Data, Predict, Scenario, Optimize, Act. "Dogs Play Softly On Asphalt." Notice the predictive model (step 2) happens before the optimization (step 4).

Key Technologies And Techniques

Prescriptive analytics leverages several advanced technologies to transform data into actionable recommendations:

Technology	What It Does
Machine Learning and AI	Algorithms that learn from data patterns to make predictions and improve over time without explicit programming.
Optimization Algorithms	Mathematical techniques that find the best solution among alternatives, considering constraints and objectives.
Simulation Modeling	Creating virtual scenarios to test different strategies and understand potential outcomes.
Natural Language Processing	Enabling systems to understand and communicate recommendations in human language.
Real-time Analytics	Processing data as it is created to provide immediate insights and recommendations.

Memory Aid

M-O-S-N-R: Machine learning, Optimization, Simulation, NLP, Real-time. "My Owl Sings Nice Rhymes."

Business Applications By Industry

Industry	Applications
Healthcare	Patient treatment optimization, resource allocation, staff scheduling, and personalized medicine.
Supply Chain	Inventory optimization, logistics routing, demand forecasting, and supplier selection.
Financial Services	Portfolio optimization, risk management, fraud detection, and algorithmic trading.
Manufacturing	Production scheduling, quality control, predictive maintenance, and resource allocation.
Energy and Utilities	Grid optimization, energy trading, demand response, and renewable integration.

Memory Aid

H-S-F-M-E: Healthcare, Supply chain, Financial, Manufacturing, Energy. Each industry gets its own flavor of "optimize something."

Enhancing Decision Making

Prescriptive analytics transforms organizational decision making from intuition-based to data-driven.

Benefit	Explanation
Reduced Decision Complexity	Simplifies complex decisions by evaluating numerous variables and constraints simultaneously.
Increased Decision Speed	Enables faster decision-making through automated analysis and recommendation generation.
Improved Decision Quality	Enhances outcomes by considering more factors and scenarios than humanly possible.
Competitive Advantage	Organizations using prescriptive analytics gain an edge through optimized resource allocation and strategic planning.

The result: better decisions, faster, across all levels of the organization.

Memory Aid

R-I-I-C: Reduced complexity, Increased speed, Improved quality, Competitive advantage. All four benefits start with a word meaning "better than before."

Implementation Challenges And Best Practices

Successfully implementing prescriptive analytics requires addressing several key challenges, each paired with its solution on the slide:

Challenge	Solution
Data Integration: combining data from disparate sources with varying formats and quality.	Implement robust data governance and ETL processes.
Skill Gaps: finding talent with expertise in both analytics and domain knowledge.	Invest in training programs and cross-functional teams.
Organizational Change: shifting from intuition-based to data-driven decision making culture.	Start with high-impact use cases and demonstrate ROI.
Technology Selection: choosing the right tools and platforms for specific business needs.	Begin with pilot projects to evaluate technology fit.

Memory Aid

Pair each challenge with its fix: Data -> Governance/ETL, Skills -> Training, Change -> High-impact pilots and ROI, Tech choice -> Pilot projects. The solution to "people don't trust the data-driven culture" is always to prove it works on a small scale first.

Watch Out

Four challenges, four solutions, always in the same order on the slide: **Data Integration, Skill Gaps, Organizational Change, Technology Selection**. Do not swap "training programs" (the fix for skill gaps) with "pilot projects" (the fix for technology selection); both sound like "start small" but answer different problems.

Future Trends

Trend	What It Means
Autonomous Decision Systems	Systems that not only recommend actions but can implement them with minimal human intervention.
Augmented Analytics	AI-powered tools that make advanced analytics accessible to non-technical business users.
Edge Computing	Prescriptive capabilities deployed at the edge for real-time decision making without latency.
Explainable AI	Transparent algorithms that can explain the reasoning behind their recommendations.

Prescriptive analytics represents the future of data-driven decision making. Organizations that successfully implement these capabilities will gain significant competitive advantages through optimized operations, enhanced customer experiences, and improved business outcomes.

Memory Aid

A-A-E-E: Autonomous, Augmented, Edge, Explainable. Autonomous acts for you; Augmented helps non-technical users; Edge removes latency; Explainable shows its reasoning.

One-Page Cram Sheet

Topic	Remember
Four analytics types	Descriptive (what happened), Diagnostic (why), Predictive (will happen), Prescriptive (should do)
Predictive process	Project Design, Data Sampling, Data Exploration, then a cycle of Data Modification, Model Development, Model Validation
Regression equation	$y = \beta x + \alpha + \epsilon$
Sales example	$y = \text{Sales}$, $x = \text{Ad Spending}$, $\beta = \text{Sales Sensitivity}$, $\alpha = \text{Baseline Sales}$
Multiple regression	$y = \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n + \alpha + \epsilon$
Dataset split	Original -> Training + Testing; Training -> Training + Validation
R Square	Goodness-of-fit, percent of variation explained; here 0.8937 = 89.37 percent
Adjusted R Square	Corrects R Square for number of predictors; use it to compare models
ANOVA SS	Regression SS (explained) + Residual SS (unexplained) = Total SS
ANOVA df, n=140, 3 predictors	Regression 3, Residual 136, Total 139
ANOVA MS	SS divided by its df; used only to compute F
Significance F	p-value of the whole regression; small means at least one predictor matters
When p is low	H0 must go (reject the null hypothesis)
t-test per predictor	$p < 0.05$ significant; here TV and Radio significant, Newspaper is not
Coefficient reading	Holding the other predictors constant
p26 vs p28 model	p26 t-test table keeps all 3 predictors; p28 fitted model drops newspaper and rounds (2.98, 0.047, 0.178)
MAE / MSE / RMSE / MAPE	Average absolute error / average squared error (squared units) / root of MSE (original units) / absolute error as a percent of actual Y
Prescriptive analytics	"What should we do?"; most advanced, most complex analytics stage
Five-step process	Data Collection, Predictive Modeling, Scenario Analysis, Optimization, Action Recommendation
Key technologies	Machine Learning/AI, Optimization Algorithms, Simulation, NLP, Real-time Analytics
Challenge -> Solution	Data Integration -> governance/ETL; Skill Gaps -> training; Organizational Change -> high-impact pilots/ROI; Technology Selection -> pilot projects
Future trends	Autonomous Decision Systems, Augmented Analytics, Edge Computing, Explainable AI